

Chapter 1

Homological Algebra

1.1 Derived and Triangulated Categories

Now, we give a rapid tour of the required homological algebra background.

Definition 1.1.1. Let $\mathcal{A}$ be an abelian category.

- Denote $\text{Kom}(\mathcal{A})$ the **category of chain complexes** valued in $\mathcal{A}$, with chain maps as morphisms.
- Denote $K(\mathcal{A})$ the **homotopy category** of $\mathcal{A}$, which has the same objects as $\text{Kom}(\mathcal{A})$, but morphisms are quotiented by chain homotopy.
- Denote $D(\mathcal{A})$ the **derived category** of $\mathcal{A}$, which is $K(\mathcal{A})[\text{q-isos}^{-1}]$, i.e., the category with the same objects as $K(\mathcal{A})$ but with formal inverses to quasi-isomorphisms adjoined.

Remark 1.1.2. A few remarks are in place:

- We will always use cohomology indexing for chain complexes.
- We can put superscripts $+$, $-$, b on Kom , K or D to denote the category where we only have objects which are chain complexes with cohomology vanishing sufficiently far on the left, right, or both.

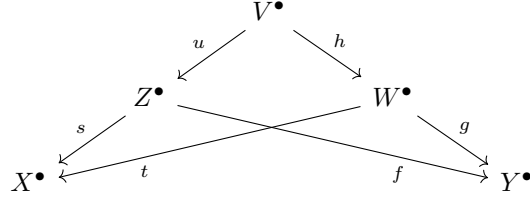
1.1.1 Morphisms in the Derived Category

In general, it is difficult to describe the morphisms of a category where we formally inverted morphisms (called **localization**, in analogy to the ring setting). However, in $K(\mathcal{A})$, the class of quasi-isomorphisms form a **localizing class** (I won't include the definition here) which makes the following true.

Definition 1.1.3. A **roof** in $D(\mathcal{A})$ is a diagram of the form

$$\begin{array}{ccc} & Z^\bullet & \\ s \swarrow & & \searrow f \\ X^\bullet & & Y^\bullet \end{array}$$

for s a quasi-isomorphism, and f a chain map (or rather both should be homotopy classes of maps), which should be thought of as $f \circ s^{-1} = \frac{f}{s}$. The roofs $X^\bullet \xleftarrow{s} Z^\bullet \xrightarrow{f} Y^\bullet$ and $X^\bullet \xleftarrow{t} W^\bullet \xrightarrow{g} Y^\bullet$ are **equivalent** if there exists a “common denominator” roof, i.e., there is a roof $Z^\bullet \xleftarrow{u} V^\bullet \xrightarrow{h} W^\bullet$ make the diagram commute (in $K(\mathcal{A})$, so up to homotopy)



which will at least imply $f \circ s^{-1} = g \circ t^{-1}$, since

$$f \circ s^{-1} = f \circ u \circ u^{-1} \circ s^{-1} = g \circ h \circ u^{-1} \circ s^{-1}$$

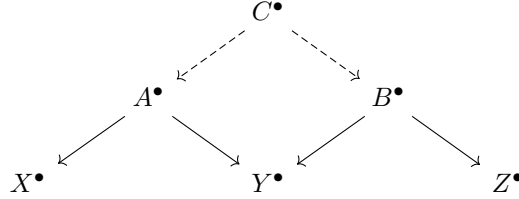
(the long route), and to get $g \circ t^{-1}$ to be the long route, we first notice

$$\begin{aligned} s \circ u &= t \circ h \\ t^{-1} &= h \circ s^{-1} \circ u^{-1} \end{aligned}$$

so with this identity

$$g \circ t^{-1} = g \circ h \circ s^{-1} \circ u^{-1}.$$

We can compose roofs $X^\bullet \leftarrow A^\bullet \rightarrow Y^\bullet$ and $Y^\bullet \leftarrow B^\bullet \rightarrow Z^\bullet$ by putting a roof over both of them.



This is always be possible, exercise for the reader. You should do this by studying cones/distinguished triangles.

Theorem 1.1.4.

$$\mathrm{Hom}_{D(\mathcal{A})}(X^\bullet, Y^\bullet) = \{\text{roofs } X^\bullet \text{ to } Y^\bullet\} / \text{equivalence},$$

i.e., all morphisms in the derived category are roofs.

In some cases, we can describe the morphisms even more explicitly. There is a functor $\mathcal{A} \rightarrow D(\mathcal{A})$ by sending an object to a degree 0 complex $A \mapsto (\cdots \rightarrow 0 \rightarrow A \rightarrow 0 \rightarrow \cdots)$. We will often identify $A \in \mathcal{A}$ with its degree 0 complex. We can then ask the following:

- Is $\mathcal{A} \rightarrow D(\mathcal{A})$ full? faithful? That is for $A, B \in \mathcal{A}$, can we describe

$$\mathrm{Hom}_{D(\mathcal{A})}(A, B)$$

in relation to $\mathrm{Hom}_{\mathcal{A}}(A, B)$?

- Can we understand $\mathrm{Hom}_{D(\mathcal{A})}(A, B[i])$?

For the second question, at least when $i > 0$, we have the following strategy for writing down morphisms. Take an exact sequence

$$0 \rightarrow B \rightarrow K^{-i+1} \rightarrow \cdots \rightarrow K^0 \rightarrow A \rightarrow 0,$$

which we denote by $K^\bullet$. In this case, we have the morphisms of chain complexes

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots & \longrightarrow & 0 & \longrightarrow & A & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & & & & & & & & & \uparrow & & & & \\ 0 & \longrightarrow & B & \longrightarrow & K^{-i+1} & \longrightarrow & \cdots & \longrightarrow & K^{-1} & \longrightarrow & K^0 & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & \downarrow & & & & & & & & & & & & \\ 0 & \longrightarrow & B & \longrightarrow & 0 & \longrightarrow & \cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots \end{array}$$

where the top morphism is a quasi-isomorphism (so has a formal inverse). Denote the middle complex by $\tilde{K}$, so that in total we get a morphism $A \rightarrow B[i]$ which is a roof.

$$\begin{array}{ccc} & \tilde{K} & \\ \swarrow & & \searrow \\ A & & B[i] \end{array}$$

Denote this morphism by $y(K^\bullet)$. We can then ask whether all morphisms $A \rightarrow B[i]$ arise in this way, and when are two such morphisms equivalent.

These questions are then answered by the following proposition.

Proposition 1.1.5. (a) $\text{Hom}_{D^*(\mathcal{A})}(X, Y[i]) = 0$ for $i < 0$

(b) $\text{Hom}_{D^*(\mathcal{A})}(X, Y) = \text{Hom}_{\mathcal{A}}(X, Y)$

(c) Any element of $\text{Hom}_{D^*(\mathcal{A})}(X, Y[i])$ is of the form $y(K^\bullet)$, and the composition corresponds to concatenating extensions.

Proof of (a). Let $\varphi : X \rightarrow Y[-i]$ for $i > 0$ be written as a roof $X \xleftarrow{s} K^\bullet \xrightarrow{f} Y[-i]$. To show $\varphi = 0$, we □

We also have the following theorem to help us understand morphisms: TODO

Proposition 1.1.6. $K^+(I) \rightarrow D^+(\mathcal{A})$ and $K^-(P) \rightarrow D^-(\mathcal{A})$ form equivalence of categories (assuming enough projectives/injectives). Consequently,

$$\text{Hom}_{K(\mathcal{A})}(X^\bullet, Y^\bullet) \cong \text{Hom}_{D(\mathcal{A})}(X^\bullet, Y^\bullet)$$

when either $Y^\bullet \in I$ or $X^\bullet \in P$.

Proof. both of these would be good to understand!!! □

1.1.2 Triangles

The derived category is usually no longer abelian, but it does have some extra structure. It's still additive (add roofs by expressing both as $f \circ s^{-1}, g \circ s^{-1}$ for some s , and then $(f + g) \circ s^{-1}$ is the sum), it still has the (additive) shift functor $X^\bullet \mapsto X^\bullet[1]$, and it has a notion of “distinguished triangles”, which somewhat analogous to exact sequences.

Recall that a short exact sequence

$$0 \rightarrow A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet \rightarrow 0 \tag{*}$$

of chain complexes in $\text{Kom}(\mathcal{A})$ induces a long exact sequence in cohomology.

$$H^n(A^\bullet) \xrightarrow{H^n(f)} H^n(B^\bullet) \xrightarrow{H^n(g)} H^n(C^\bullet) \xrightarrow{\delta^n} H^{n+1}(A^\bullet) = H^n(A^\bullet[1]),$$

where the maps $H^n(f), H^n(g)$ are directly induced by f, g , but δ^n is maybe more mysterious. In any case, the LES might be drawn as a spiraling triangular helix.

$$\begin{array}{ccccc}
& & & & \cdots \\
& & & \nearrow & \\
H^n(B[1]) & & & & \\
\searrow & \nwarrow & & \nearrow & \\
H^n(C[1]) & & & H^n(A[1]) & \\
& \nwarrow & \nearrow & & \\
H^n(B) & & & & \\
\searrow & \nwarrow & & \nearrow & \\
H^n(C) & & & H^n(A) & \\
& \nwarrow & \nearrow & & \\
& \cdots & & &
\end{array}$$

One thing the derived category does for you is it makes it so δ^n is actually induced by a morphism $C \rightarrow A[1]$, so that this spiral is true not just on the level of cohomology, but on the level of the derived category.

More precisely, it is true that the short exact sequence $(*)$ is quasi-isomorphic to a short exact sequence

$$0 \rightarrow A^\bullet \rightarrow \text{Cyl}(f) \rightarrow C(f) \rightarrow 0,$$

where $\text{Cyl}(f), C(f)$ are the **mapping cylinder/mapping cone** respectively, analogous to topology, but they have a chain complex definition. In some sense, they are defined exactly the same as in topology—define the path object chain complex $I^\bullet$ by $I^{-1} = \mathbb{Z}e$, $I^0 = \mathbb{Z}a \oplus \mathbb{Z}b$, and $d^{-1}(e) = b - a$. Then, $\text{Cyl}(f)$ is the pushout of a diagram $A^\bullet \otimes I^\bullet \leftarrow A^\bullet \rightarrow B^\bullet$, and $C(f)$ is a quotient of $\text{Cyl}(f)$.

Also, like topology, there is a map

$$C(f) \xrightarrow{\delta} SA^\bullet := A^\bullet[1]$$

where we $SA^\bullet$ is the suspension (this map is “contracting $B^\bullet$ ”). This map induces the connecting map in the long exact sequence in cohomology, i.e.,

$$H^n(C(f)) \xrightarrow{H^n(\delta)} H^n(A^\bullet[1]) = H^{n+1}(A^\bullet)$$

is the connecting map. We will call a sequence of the form

$$A^\bullet \rightarrow \text{Cyl}(f) \rightarrow C(f) \rightarrow A^\bullet[1]$$

(or anything quasi-isomorphic to it) a **distinguished triangle** in $D(\mathcal{A})$. What we just discussed shows that any short exact sequence in $\text{Kom}(\mathcal{A})$ gives a distinguished triangle in $D(\mathcal{A})$.

Just as a short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ tells us that B is C extended by A , a distinguished triangle also tells us this sort of gluing information but with added symmetry. We axiomatize the formal structure of the derived category with the following notion.

Definition 1.1.7. A **triangulated category** Δ is an additive category with

- An additive automorphism $(-)[1] : \Delta \rightarrow \Delta$ called the **shift/translation/suspension functor** (some authors relax to auto-equivalence, but this is tricky).

- A collection of triangles

$$\{A^\bullet \rightarrow B^\bullet \rightarrow C^\bullet \rightarrow A^\bullet[1]\}$$

called distinguished triangles

such that the following axioms hold:

- **TR1:**

- Every triangle isomorphic to a distinguished triangle is itself a distinguished triangle.
- The triangle

$$X \xrightarrow{\text{id}_X} X \rightarrow 0 \rightarrow X[1]$$

is a distinguished triangle.

- Every morphism $f : X \rightarrow Y$ can be completed to a distinguished triangle

$$X \xrightarrow{f} Y \rightarrow C(f) \rightarrow X[1],$$

where $C(f)$ is called the **cone** of f . The other axioms will imply $C(f)$ is unique up to (non-unique) isomorphism.

- **TR2:** a triangle

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

is distinguished iff

$$Y \xrightarrow{g} Z \xrightarrow{h} X[1] \xrightarrow{-T(f)} Y[1]$$

is distinguished (we can “rotate” distinguished triangles, giving us the spiraling triangular helix we drew earlier).

- **TR3:** Given a diagram of solid arrows

$$\begin{array}{ccccccc} X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \xrightarrow{h} & X[1] \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow T(\alpha) \\ X' & \xrightarrow{f'} & Y' & \xrightarrow{g'} & Z' & \xrightarrow{h'} & X'[1] \end{array}$$

where the rows are distinguished triangles and the left square commutes, then there exists $\gamma : Z \rightarrow Z'$ making the entire diagram commute.

- **TR4 (octahedral axiom):** Given three distinguished triangles of the form

$$\begin{array}{ccccccc} X & \xrightarrow{f} & Y & \xrightarrow{p} & Y/X & \longrightarrow & X[1] \\ Y & \xrightarrow{g} & Z & \xrightarrow{q} & Z/Y & \longrightarrow & Y[1] \\ X & \xrightarrow{g \circ f} & Z & \xrightarrow{r} & Z/X & \longrightarrow & X[1] \end{array}$$

(where we are denoting the cones suggestively like quotients), there exists a distinguished triangle

$$Y/X \xrightarrow{\bar{g}} Z/X \xrightarrow{\pi} Z/Y \xrightarrow{s} (Y/X)[1]$$

such that the following diagram commutes

$$\begin{array}{ccccccc} X & \xrightarrow{g \circ f} & Z & \xrightarrow{q} & Z/Y & \xrightarrow{s} & (Y/X)[1] \\ & \searrow f & \nearrow g & \searrow r & \nearrow \pi & \searrow p[1] & \\ & & Y & & Z/X & & Y[1] \\ & & \searrow p & \nearrow \bar{g} & \searrow f[1] & & \\ & & & Y/X & \longrightarrow & X[1] & \end{array}$$

Remark 1.1.8. The octahedral axiom is roughly a version of the “third isomorphism theorem”, which says $(Z/X)/(Y/X) \cong Z/Y$. The commuting diagram says something about how the projections/suspensions interact.

Remark 1.1.9. In TR3, we do *not* assert that γ is unique, just that some γ exists. In fancier words, the cone construction is not functorial.

Theorem 1.1.10. For $\mathcal{A}$ and abelian category, $D^*(\mathcal{A})$ (where $*$ $\in \{+, -, b, \}$) is a triangulated category with the shift functor and the described distinguished triangles.

We will not prove this in these notes, but I don’t believe it is particularly hard.

Now, we’re going to give ourselves a little bit of scaffolding for how to handle a triangulated category. We’re not going to be very comprehensive—go to the Stacks’ project if that’s what you want.

Proposition 1.1.11. In a distinguished triangle

$$X \xrightarrow{f} Y \xrightarrow{g} Z \rightarrow X[1],$$

$$g \circ f = 0.$$

Proof. By TR1 we have the distinguished triangle $X \xrightarrow{\text{id}_X} X \rightarrow 0 \rightarrow X[1]$. We can then apply TR3 to get the following commutative diagram

$$\begin{array}{ccccccc} X & \xrightarrow{\text{id}_X} & X & \longrightarrow & 0 & \longrightarrow & X[1] \\ \downarrow \text{id}_X & & \downarrow f & & \downarrow & & \downarrow \text{id}_{X[1]} \\ X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \longrightarrow & X[1] \end{array}$$

so indeed $g \circ f = 0$. □

This tells us, at least a little bit, distinguished triangles behave like exact sequences.

Definition 1.1.12. Let Δ a triangulated category, and $\mathcal{A}$ an abelian category. A **cohomomological functor** is a functor $H : \Delta \rightarrow \mathcal{A}$ such that every distinguished triangle

$$X \rightarrow Y \rightarrow Z \rightarrow X[1]$$

is sent to an exact sequence

$$H(X) \rightarrow H(Y) \rightarrow H(Z)$$

(and by TR2, we can continue this exact sequence in a spiral).

Example 1.1.13. When $\Delta = D^*(\mathcal{A})$, the usual cohomology functor H^n on $\text{Kom}(\mathcal{A})$ descends to a functor on $D^*(\mathcal{A})$ (almost by definition), and indeed, the cohomology functor is a cohomological functor.

Example 1.1.14. Let Δ be a triangulated category. In particular, Δ is additive, so the hom-sets are abelian groups. So, for any $W \in \Delta$, we can consider the hom functor as

$$\text{Hom}(W, -) : \Delta \longrightarrow \mathbb{Z}\text{-Mod}.$$

We claim this is a cohomological functor. Let

$$X \xrightarrow{f} Y \xrightarrow{g} Z \rightarrow X[1]$$

be distinguished, we show exactness at the middle term

$$\text{Hom}(W, X) \xrightarrow{\text{Hom}(W, f)} \text{Hom}(W, Y) \xrightarrow{\text{Hom}(W, g)} \text{Hom}(W, Z).$$

We at least have $\text{Hom}(W, g \circ f) = 0$, since $g \circ f = 0$. Now, let $y : W \rightarrow Y$ such that $g \circ y = 0$. We wish to show $y = f \circ x$. To see this, rotate with TR2, apply TR3, and rotate back to get the commutative diagram

$$\begin{array}{ccccccc} W & \xrightarrow{\text{id}_W} & W & \longrightarrow & 0 & \longrightarrow & W[1] \\ \downarrow x & & \downarrow y & & \downarrow 0 & & \downarrow x[1] \\ X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \longrightarrow & X[1] \end{array}$$

so indeed, we get exactness.

Proposition 1.1.15. Cones are unique up to isomorphism.

Proof. This will essentially be the 5-lemma. Suppose

$$X \rightarrow Y \rightarrow Z \rightarrow X[1], \quad X \rightarrow Y \rightarrow Z' \rightarrow X[1]$$

are both distinguished triangles. Then, applying TR3, we get $\gamma : Z \rightarrow Z'$ giving us a morphism between the distinguished triangles. To show γ is an isomorphism it suffices, by Yoneda, to show for all $W \in \Delta$, $\text{Hom}(W, \gamma)$ is an isomorphism. Indeed, applying the functor $\text{Hom}(W, -)$ to the diagram which is the morphism of distinguished triangles, we get

$$\begin{array}{ccccccccc} \text{Hom}(W, X) & \longrightarrow & \text{Hom}(W, Y) & \longrightarrow & \text{Hom}(W, Z) & \longrightarrow & \text{Hom}(W, X[1]) & \longrightarrow & \text{Hom}(W, Y[1]) \\ \downarrow (id_X)_* & & \downarrow (id_Y)_* & & \downarrow \gamma_* & & \downarrow (id_{X[1]})_* & & \downarrow (id_{Y[1]})_* \\ \text{Hom}(W, X) & \longrightarrow & \text{Hom}(W, Y) & \longrightarrow & \text{Hom}(W, Z') & \longrightarrow & \text{Hom}(W, X[1]) & \longrightarrow & \text{Hom}(W, Y[1]) \end{array}$$

where all vertical morphisms (except for potentially the middle one) are isomorphisms, so by the 5-lemma the middle is an isomorphism. $\square$

Proposition 1.1.16. The direct sum of distinguished triangles is distinguished, i.e., if

$$X_i \rightarrow Y_i \rightarrow Z_i \rightarrow X_i[1] \tag{S_i}$$

is distinguished for $i = \{1, 2\}$, then

$$X_1 \oplus X_2 \rightarrow Y_1 \oplus Y_2 \rightarrow Z_1 \oplus Z_2 \rightarrow X_1[1] \oplus X_2[1]$$

Proof. We at least have distinguished triangle

$$X_1 \oplus X_2 \rightarrow Y_1 \oplus Y_2 \rightarrow Z \rightarrow X_1[1] \oplus X_2[1], \tag{T}$$

and we want to show that this is isomorphic to our “desired” triangle. We have morphisms $S_i \rightarrow T$ by completing the commutative square

$$\begin{array}{ccc} X_i & \longrightarrow & Y_i \\ \downarrow & & \downarrow \\ X_1 \oplus X_2 & \longrightarrow & Y_1 \oplus Y_2 \end{array}$$

to a distinguished triangle. Now, we want to argue that $S_1 \oplus S_2 \rightarrow T$ is an isomorphism. We do this by hitting the morphism of distinguished triangles with $\text{Hom}(W, -)$. Even though $S_1 \oplus S_2$ is not a-priori distinguished, it will still turn into an exact sequence after being hit with $\text{Hom}(W, -)$, since it will be a direct sum of exact sequences $\text{Hom}(W, S_i)$. Therefore, we can finish the proof by applying the 5-lemma. $\square$

Proposition 1.1.17. The morphism $A \xrightarrow{0} B$ is completed to a distinguished triangle via

$$A \xrightarrow{0} B \xrightarrow{i} A[1] \oplus B \xrightarrow{\pi} A[1],$$

where i, π are inclusions/projections into the direct sum.

Proof. We have distinguished triangles

$$A \xrightarrow{\text{id}_A} A \rightarrow 0 \rightarrow A[1],$$

which rotates via TR3 to

$$A \rightarrow 0 \rightarrow A[1] \xrightarrow{\text{id}_{A[1]}} A[1].$$

Likewise, we have distinguished triangle

$$0 \rightarrow B \xrightarrow{\text{id}_B} B \rightarrow 0[1].$$

Their direct sum is a distinguished triangle

$$A \xrightarrow{0} B \xrightarrow{i} A[1] \oplus B \xrightarrow{\pi} A[1].$$

□

We will define/develop some helpful notation.

Definition 1.1.18. Let $A, C \in \mathcal{D}$. We call B an **extension** of A, C if there is a distinguished triangle

$$A \rightarrow B \rightarrow C \rightarrow A[1].$$

Let $E_1, \dots, E_n \in \mathcal{D}$ be some objects (resp. $\mathcal{C} \subseteq \mathcal{D}$ a full subcategory). Denote the **extension closure**

$$\langle E_1, \dots, E_n \rangle \subseteq \mathcal{D}, \quad (\text{resp. } \langle \mathcal{C} \rangle \subseteq \mathcal{D})$$

the smallest full subcategory containing $E_1, \dots, E_n$ (resp. $\mathcal{C}$) which is closed under extensions.

Definition 1.1.19. Let $\mathcal{C} \subseteq \mathcal{D}$ be a full subcategory of a triangulated category $\mathcal{D}$. Denote

$${}^\perp \mathcal{C} = \{X \in \mathcal{D} \mid \text{Hom}(X, \mathcal{C}) = 0\},$$

where by $\text{Hom}(X, \mathcal{C}) = 0$, we mean for all $C \in \mathcal{C}$, $\text{Hom}(X, C) = 0$. Similarly, define

$$\mathcal{C}^\perp = \{X \in \mathcal{D} \mid \text{Hom}(\mathcal{C}, X) = 0\}.$$

If we squint our eyes and think of Hom as some sort of non-symmetric inner product, then we are exactly looking at the (one-sided) orthogonal complements. We put the $\perp$ on the side of $\mathcal{C}$ where we should put X to get Hom with $\mathcal{C}$ to be 0.

Proposition 1.1.20. ${}^\perp \mathcal{C}$ and $\mathcal{C}^\perp$ are closed under extensions.

Proof. Let $X \rightarrow Y \rightarrow Z \rightarrow X[1]$ be a distinguished triangle with $X, Z \in {}^\perp \mathcal{C}$. We want to show $Y \in {}^\perp \mathcal{C}$, so for an arbitrary object $C \in \mathcal{C}$, we show $\text{Hom}(Y, C) = 0$.

To do this, we hit the triangle with $\text{Hom}(-, C)$ to get the exact sequence

$$\text{Hom}(X, C) \leftarrow \text{Hom}(Y, C) \leftarrow \text{Hom}(Z, C),$$

and by assumption the left and right terms are 0, so $\text{Hom}(Y, C) = 0$. This shows ${}^\perp \mathcal{C}$ is closed under extensions.

An analogous proof shows $\mathcal{C}^\perp$ is also closed under extensions. □

Corollary 1.1.21. Let $\mathcal{C}, \mathcal{C}'$ be full subcategories of $\mathcal{D}$. If $\mathcal{C}' \subseteq {}^\perp \mathcal{C}$, then $\langle \mathcal{C}' \rangle \subseteq {}^\perp \mathcal{C}$.

Proposition 1.1.22 (Verdier's Exercise/The 3×3 lemma). Assume

$$\begin{array}{ccc} X & \longrightarrow & Y \\ \downarrow & & \downarrow \\ Z & \longrightarrow & W \end{array}$$

is commutative. Then, there exists an object U and morphisms making all rows and columns distinguished triangles.

$$\begin{array}{ccccccc}
X & \longrightarrow & Y & \longrightarrow & Y/X & \longrightarrow & X[1] \\
\downarrow & & \downarrow & & \downarrow & & \downarrow \\
Z & \longrightarrow & W & \longrightarrow & W/Z & \longrightarrow & Z[1] \\
\downarrow & & \downarrow & & \downarrow & & \downarrow \\
Z/X & \dashrightarrow & W/Y & \dashrightarrow & U & \dashrightarrow & (Z/X)[1] \\
\downarrow & & \downarrow & & \downarrow & & \downarrow \\
X[1] & \longrightarrow & Y[1] & \longrightarrow & (Y/X)[1] & \longrightarrow & X[2]
\end{array}$$

Proof Sketch. We get $Y/X \rightarrow W/Z$ by TR1 and TR3, likewise for $Z/X \rightarrow W/Y$ (i.e., we get the diagram without the bottom 2×2 square, with rows and columns distinguished). Following TR1, we get two “options” for U , which we will denote respectively

$$(W/Z)/(Y/X), \quad (W/Y)/(Z/X).$$

Heuristically, if cross multiply we get the same formal fractions which gives us some hope they are isomorphic and that the squares commute. □

1.2 t-structures

The derived category $D(\mathcal{A})$ of an abelian category $\mathcal{A}$ comes with even more structure called a t-structure—essentially this is the inclusion of $\mathcal{A} \hookrightarrow D(\mathcal{A})$ as degree 0 complexes.

Remark 1.2.1. We can understand $\mathcal{A} \hookrightarrow D(\mathcal{A})$ as the codomain category of H^0 . In fact, we can understand H^0 as *truncation*. Define $D(\mathcal{A})^{\leq 0}$ as the full subcategory of complexes $A^\bullet$ with $H^n(A^\bullet) = 0$ for $n > 0$, and $D(\mathcal{A})^{\geq 0}$ as those with $H^n(A^\bullet) = 0$ for $n < 0$.

Now, define the left truncation functor

$$\begin{aligned}
\tau_{\geq 0} : D(\mathcal{A}) &\longrightarrow D(\mathcal{A})^{\geq 0} \\
\left(\cdots \rightarrow A^{-1} \xrightarrow{d^0} A^0 \xrightarrow{d^1} A^1 \rightarrow \cdots \right) &\longmapsto (\cdots \rightarrow 0 \rightarrow 0 \rightarrow \operatorname{coker} d^0 \rightarrow A^1 \rightarrow \cdots) \\
&\simeq (\cdots \rightarrow 0 \rightarrow \operatorname{im} d^{-1} \hookrightarrow A^0 \rightarrow A^1 \rightarrow \cdots)
\end{aligned}$$

and the right truncation functor

$$\begin{aligned}
\tau_{\leq 0} : D(\mathcal{A}) &\longrightarrow D(\mathcal{A})^{\leq 0} \\
\left(\cdots \rightarrow A^{-1} \xrightarrow{d^0} A^0 \xrightarrow{d^1} A^1 \rightarrow \cdots \right) &\longmapsto (\cdots \rightarrow A^{-1} \rightarrow \ker d^0 \rightarrow 0 \rightarrow 0 \rightarrow \cdots) \\
&\simeq (\cdots \rightarrow A^{-1} \rightarrow A^0 \twoheadrightarrow \operatorname{im} d^1 \rightarrow 0 \rightarrow \cdots)
\end{aligned}$$

where these functors really are defined on the level of derived categories, i.e., quasi-isomorphic complexes give quasi-isomorphic truncations. Then, one can check for $i : \mathcal{A} \rightarrow D(\mathcal{A})$ the inclusion of degree 0 complexes,

$$i \circ H^0 \cong \tau_{\leq 0} \circ \tau_{\geq 0} \cong \tau_{\geq 0} \circ \tau_{\leq 0},$$

where the compositions commuting is the fact that for a complex $A \xrightarrow{f} B \xrightarrow{g} C$

$$\operatorname{coker}(A \xrightarrow{f'} \ker g) \cong \ker g / \operatorname{im} f \cong \ker(\operatorname{coker} f \xrightarrow{\bar{g}} C).$$

So, we'll see a t-structure as a way to give generalized cohomology. Other cohomologies besides H^0 are gotten just by shifting.

Remark 1.2.2. We can then refine our above ideas that cohomology is a way of giving objects a filtration in the derived category. Given an object $A^\bullet \in D(\mathcal{A})$, we have the **cohomology filtration**

$$0 \rightarrow \cdots \rightarrow \tau_{\leq n-1} A^\bullet \rightarrow \tau_{\leq n} A^\bullet \rightarrow \cdots \rightarrow A$$

where we have the short exact sequence

$$0 \rightarrow \tau_{\leq n-1} A^\bullet \rightarrow \tau_{\leq n} A^\bullet \rightarrow (\cdots \rightarrow 0 \rightarrow A^{n-1} / \ker d^n \rightarrow \ker d^{n+1} \rightarrow 0 \rightarrow \cdots) \rightarrow 0,$$

but in the derived category

$$(\cdots \rightarrow 0 \rightarrow A^{n-1} / \ker d^n \rightarrow \ker d^{n+1} \rightarrow 0 \rightarrow \cdots) \simeq (\cdots \rightarrow 0 \rightarrow 0 \rightarrow H^n(A^\bullet) \rightarrow 0 \rightarrow \cdots),$$

since there is a map left to right inducing isomorphisms of cohomology. So, in the derived category we have the triangle

$$\tau_{\leq n-1} A^\bullet \rightarrow \tau_{\leq n} A^\bullet \rightarrow H^n(A^\bullet)[-n] \rightarrow \tau_{\leq n-1} A^\bullet[1],$$

where the $-n$ shift is to place the cohomology at degree n . If we think of cones as a derived category analogy of quotients/kernels, we should view this as saying that $\tau_{\leq n} A^\bullet$ is built by gluing $H^n(A^\bullet)[-n]$ to $\tau_{\leq n-1} A^\bullet$.

In this way, the abelian category $\mathcal{A} \hookrightarrow D(\mathcal{A})$ is important because shifts of it (the cohomologies of a complex) are used to build the entire derived category by gluing. We will give two related definitions, which are equivalent in certain setups explained below.

1.2.1 Basics

This first definition defines a t-structure by defining the image of the truncation functors—by specifying $\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0}$, we will later define the truncations as adjoints to the inclusions of these categories (and the adjoints will exist).

Definition 1.2.3. Let $\mathcal{D}$ be a triangulated category. A **t-structure** on $\mathcal{D}$ is a pair $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ of strictly full subcategories (strictly=closed under isomorphisms), which satisfying some axioms. Denote $\mathcal{D}^{\leq n} := \mathcal{D}^{\leq 0}[-n]$, $\mathcal{D}^{\geq n} := \mathcal{D}^{\geq 0}[n]$, and require

- (i) $\text{Hom}_{\mathcal{D}}(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 1}) = 0$.
- (ii) If $X \in \mathcal{D}^{\leq 0}$, then $X[1] \in \mathcal{D}^{\leq 0}$. If $Y \in \mathcal{D}^{\geq 0}$, then $Y[-1] \in \mathcal{D}^{\geq 0}$.
- (iii) For $A \in \mathcal{D}$, there exists a distinguished triangle

$$X \rightarrow A \rightarrow Y \rightarrow X[1]$$

such that $X \in \mathcal{D}^{\leq 0}$ and $Y \in \mathcal{D}^{\geq 1}$. These will be unique up to isomorphism, and we denote $\tau^{\leq 0} A := X$ and $\tau^{\geq 1} A := Y$

In particular, (i) implies $\mathcal{D}^{\leq 0} \cap \mathcal{D}^{\geq 1} = 0$, because otherwise a nonzero object E would have the nonzero morphism id_E .

The t-structure is said to be **non-degenerate** if $\bigcap_{n \in \mathbb{N}} \mathcal{D}^{\leq n} = \bigcap_{n \in \mathbb{N}} \mathcal{D}^{\geq n} = \{0\}$. It is said to be **bounded** if for all $E \in \mathcal{D}$, there exists $n \in \mathbb{N}$ such that $E \in \mathcal{D}^{\leq n} \cap \mathcal{D}^{\geq -n}$. In particular, bounded t-structures are non-degenerate because a nonzero $E \in \mathcal{D}^{\leq n} \cap \mathcal{D}^{\geq -n}$ will *not* be in $\mathcal{D}^{\geq n+1}$ nor $\mathcal{D}^{\leq n-1}$.

We denote $\tau^{\leq n} A = \tau^{\leq 0}(A[n])[-n]$ and $\tau^{\geq n} A = \tau^{\geq 1}(A[n-1])[-n+1]$, called the **truncation functors** (we will soon prove they are well-defined and functorial).

The **heart** of the t-structure is $\mathcal{D}^{\leq 0} \cap \mathcal{D}^{\geq 0}$, denoted $\mathcal{D}^\heartsuit$. We will later see that the heart of a t-structure is abelian.

For $A \in \mathcal{D}$, we denote $H_{\mathcal{P}}^n(A) := (\tau^{\leq n} \tau^{\geq n} A)[n] \in \mathcal{D}^\vee$, called the **cohomology** of A with respect to the t-structure. This name makes sense, because we will later prove that

$$H_{\mathcal{P}}^0 : \mathcal{D} \rightarrow \mathcal{D}^\vee$$

is a cohomological functor.

Proposition 1.2.4. Let $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ be a t-structure. Then, $\tau^{\leq 0}, \tau^{\geq 1}$ (as well as their translates) are well-defined functors, and they form adjoint pairs

$$(i^{\leq 0}, \tau^{\leq 0}), \quad (\tau^{\geq 0}, i^{\geq 0})$$

Proof. To show the truncations are well-defined functors, we show that given $f : A \rightarrow A'$ and distinguished triangles

$$X \rightarrow A \rightarrow Y \rightarrow X[1], \quad X' \rightarrow A' \rightarrow Y' \rightarrow X'[1]$$

with $X, X' \in \mathcal{D}^{\leq 0}$ and $Y, Y' \in \mathcal{D}^{\geq 1}$, then we get a unique morphism of the distinguished triangles.

In particular, supposing we have shown this, we can apply this to $f = \text{id}_A$ twice to get a diagram

$$\begin{array}{ccccccc} X & \longrightarrow & A & \longrightarrow & Y & \longrightarrow & X[1] \\ \downarrow & \searrow & \downarrow & \searrow & \downarrow & \searrow & \downarrow \\ X' & \longrightarrow & A & \longrightarrow & Y' & \longrightarrow & X'[1] \\ \downarrow & \searrow & \downarrow & \searrow & \downarrow & \searrow & \downarrow \\ X & \longrightarrow & A & \longrightarrow & Y & \longrightarrow & X[1] \end{array}$$

and by uniqueness of the morphisms “extending” id_A , we are forced to have the composition be identity. This argument eventually tells us that the triangle given by the t-structure is unique up to isomorphism.

So, we just have to show this fact. We start by completing the diagram.

$$\begin{array}{ccc} X & \longrightarrow & A \\ \downarrow & & \downarrow f \\ X' & \longrightarrow & A' \end{array}$$

Since $\text{Hom}(X, -)$ is a cohomological functor, we have an exact sequence

$$\text{Hom}(X, Y'[-1]) \rightarrow \text{Hom}(X, X') \rightarrow \text{Hom}(X, A') \rightarrow \text{Hom}(X, Y'),$$

and we want the middle map an isomorphism, so that the composition $X \rightarrow A \xrightarrow{f} A'$ is the pushforward of a unique $X \rightarrow X'$. Now, this follows from the axioms—we have $Y', Y'[-1] \in \mathcal{D}^{\geq 0}$ whereas $X \in \mathcal{D}^{\leq 0}$, so the left and right terms in the exact sequence vanish. A similar argument gives a unique map $Y \rightarrow Y'$ making a square commute, and these morphisms must assemble into a map of distinguished triangles by TR3.

Now, we can indeed define the truncation functor $\tau^{\leq 0}$ just by picking for each A some isomorphism representative of X . We use a similar argument for $\tau^{\geq 1}$.

Finally, we say some words about the first adjunction. Given $X \in \mathcal{D}^{\leq 0}$ and $A \in \mathcal{D}$. Then, a morphism $X \rightarrow A$ in $\mathcal{D}$ gives a unique morphism $\tau^{\leq 0} X \rightarrow A$ and vice versa by the argument described before. This gives the bijection between the hom sets—one is left to check it is natural, and to work out the other adjunction. $\square$

1.2.2 Bounded t-structures

Now, we give an alternative definition of a bounded t-structure, which emphasizes the cohomology filtration.

Definition 1.2.5. A **bounded t-structure** is the data of a strictly full additive subcategory $\mathcal{A}^\# \leq \mathcal{D}$ such that

- (i) For $k_1 > k_2$, we have $\text{Hom}_{\mathcal{D}}(\mathcal{A}^\#[k_1], \mathcal{A}^\#[k_2]) = 0$ (“no negative Ext”).
- (ii) For every object E in $\mathcal{D}$ there are integers $k_1 > k_2 > \dots > k_n$ and a sequence (the top row)

$$0 = E^0 \longrightarrow E^1 \longrightarrow E^2 \rightarrow \dots \rightarrow E^{n-1} \longrightarrow E^n = E$$

where the cones over successive morphisms A^i are in $\mathcal{A}^\#[k_i]$. We will again call this sequence **t-structure cohomology filtration**.

Example 1.2.6. Given a t-structure cohomology filtration on E as above, we get a t-structure cohomology filtration on E/E^1 (the cone over $E^1 \rightarrow E$) by “quotienting” all the way through. That is, if we have the distinguished triangle

$$E^{i-1} \rightarrow E^i \rightarrow A^i \rightarrow E^{i-1}[1],$$

we should temporarily denote $E^i/E^{i-1} := A^i$, and write out the distinguished triangles (neglecting to write the third term)

$$\begin{aligned} E^1 &\rightarrow E^{i-1} \rightarrow E^{i-1}/E^1 \\ E^1 &\rightarrow E^i \rightarrow E^i/E^1, \end{aligned}$$

where $E^1 \rightarrow E^1$ is exactly the composite of $E^1 \rightarrow E^{i-1}$ and $E^{i-1} \rightarrow E^i$ (from the original distinguished triangle in the filtration). So, we are in the position to apply the octahedral axiom to get a distinguished triangle

$$E^{i-1}/E^1 \rightarrow E^i/E^1 \rightarrow E^i/E^{i-1} = A^i,$$

so indeed we get a sequence of morphisms $E^{i-1}/E^1 \rightarrow E^i/E^1$ which filters E using the same cones A^i . We did, however, throw away $E^1 = A^1$. We will use this construction to do induction on cohomology filtrations.

Proposition 1.2.7. The t-structure cohomology filtration, if we require all $A_i \neq 0$, is unique up to isomorphism.

Proof. Suppose

$$0 = F^0 \longrightarrow F^1 \longrightarrow F^2 \rightarrow \dots \rightarrow F^{m-1} \longrightarrow F^m = E$$

is another cohomology filtration, with $B_i \in \mathcal{A}^\#[\ell_i]$. Our strategy is to “pullback” the map $E \xrightarrow{\text{id}_E} E$ to the earliest respective objects in the filtration, then quotient out, and repeat.

Assume without loss of generality that $k_1 \geq \ell_1$. We can factor $E^1 \rightarrow E$ through F^1 , since we have distinguished triangle

$$F^1 \rightarrow E \rightarrow E/F^1 \rightarrow F^1[1], \quad (*)$$

where by E/F^1 has a t-structure cohomology filtration by only $B^2, \dots, B^m$, so

$$E/F^1 \in \langle \mathcal{A}^\#[\ell_2], \dots, \mathcal{A}^\#[\ell_m] \rangle \subseteq \mathcal{A}^\#[k_1]^\perp,$$

so in particular the composition $E^1 \rightarrow E \rightarrow E/F^1$ is zero. By applying $\text{Hom}(E^1, \cdot)$ to $(*)$, we get exact sequence

$$\text{Hom}(E^1, F^1) \rightarrow \text{Hom}(E^1, E) \rightarrow \text{Hom}(E^1, E/F^1),$$

so $E^1 \rightarrow E \rightarrow E/F^1$ being 0 implies we can pull back $E^1 \rightarrow E$ to $E^1 \rightarrow F^1$.

Now, if $k_1 > \ell_1$ strictly, then E_1 even factors through $E^0 = 0$, so $E^1 \rightarrow E$ is 0. Even better, we will get $E^1 \xrightarrow{\text{id}_{E^1}} E^1$ is zero, which would be a contradiction because $E^1 = A^1 \neq 0$. This is because we have the exact sequence

$$\text{Hom}(E^1, (E/E^1)[-1]) \rightarrow \text{Hom}(E^1, E^1) \rightarrow \text{Hom}(E^1, E),$$

with lefthand side 0, since

$$E/E^1 \in \langle \mathcal{A}^\sharp[k_2], \dots, \mathcal{A}^\sharp[k_n] \rangle \subseteq \mathcal{A}^\sharp[k_1]^\perp.$$

Therefore, we indeed have $k_1 = \ell_1$.

Now, we argue that $E^1 \rightarrow F^1$ is an isomorphism, by considering its cone, which we denote F^1/E^1 . We have the following distinguished triangles (neglecting to write the last term).

$$\begin{aligned} E^1 &\rightarrow F^1 \rightarrow F^1/E^1 \\ E^1 &\rightarrow E \rightarrow E/E^1 \\ F^1 &\rightarrow E \rightarrow E/F^1, \end{aligned}$$

where $E^1 \rightarrow E$ is the composite of $E^1 \rightarrow F^1 \rightarrow E$, so we are exactly in the position to apply the octahedral axiom to get the distinguished triangle

$$F^1/E^1 \rightarrow E/E^1 \rightarrow E/F^1,$$

but $F^1/E^1 \in \langle \mathcal{A}^\sharp[k_1], \mathcal{A}^\sharp[k_1 + 1] \rangle$ while $E/E^1 \in \langle \mathcal{A}^\sharp[k_2], \dots, \mathcal{A}^\sharp[k_n] \rangle$, and as usual there are no “negative Exts”, so $F^1/E^1 \rightarrow E/E^1$ is the zero morphism, so by Proposition 1.1.17,

$$E/F^1 \cong E/E^1 \oplus (F^1/E^1)[1],$$

but, if we assume $E^1 \rightarrow F^1$ is not an isomorphism, then $F^1/E^1 \neq 0$, so we get a nonzero morphism via the inclusion

$$(F^1/E^1)[1] \rightarrow E/E^1 \oplus (F^1/E^1)[1] \cong E/F^1,$$

which violates

$$E/F^1 \in \langle \mathcal{A}^\sharp[\ell_2], \dots, \mathcal{A}^\sharp[\ell_m] \rangle \subseteq \langle \mathcal{A}^\sharp[k_1], \mathcal{A}^\sharp[k_1 + 1] \rangle^\perp.$$

So, $E^1 \rightarrow F^1$ is an isomorphism, so we get $E/E^1 \rightarrow E/F^1$ an isomorphism via the 5-lemma from the following diagram.

$$\begin{array}{ccccccccc} E^1 & \longrightarrow & E & \longrightarrow & E/E^1 & \longrightarrow & E^1[1] & \longrightarrow & E[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ F^1 & \longrightarrow & E & \longrightarrow & E/F^1 & \longrightarrow & F^1[1] & \longrightarrow & E[1] \end{array}$$

Now, the quotiented filtrations E^i/E^1 , F^i/F^1 of $E/E^1 \cong E/F^1$ have strictly smaller n, m , so by induction they are isomorphic. To finish the argument, one should argue that the isomorphism of the quotiented filtrations lifts to an isomorphism of the original filtrations. $\square$

This next proposition is Lemma 3.2 in Bridgeland’s original paper. He says it’s “a good exercise in manipulating the definitions”.

Proposition 1.2.8. Let $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ give a t-structure, such that for all $E \in \mathcal{D}$, there exists $n \in \mathbb{N}$ with $E \in \mathcal{D}^{\leq n} \cap \mathcal{D}^{\geq -n}$. Then, $\mathcal{A}^\# := \mathcal{D}^\heartsuit$ is a bounded t-structure in the other sense.

Conversely, given a bounded t-structure specified by $\mathcal{A}^\#$, define

$$\mathcal{D}^{\leq 0} = \{E \in \mathcal{D} \mid \text{all } A_i \neq 0 \text{ has } k_i \geq 0 \text{ in the t-structure cohomology filtration}\},$$

$$\mathcal{D}^{\geq 0} = \{E \in \mathcal{D} \mid \text{all } A_i \neq 0 \text{ has } k_i \leq 0 \text{ in the t-structure cohomology filtration}\},$$

then, $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ forms a t-structure such that $\mathcal{D}^\heartsuit = \mathcal{A}^\#$ and for all $E \in \mathcal{D}$, there exists $n \in \mathbb{N}$ with $E \in \mathcal{D}^{\leq n} \cap \mathcal{D}^{\geq -n}$.

Proof. We show that for $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ as in the first definition, $\mathcal{A}^\# := \mathcal{D}^{\leq 0} \cap \mathcal{D}^{\geq 0}$ satisfies the second definition. It's clear that

$$\text{Hom}_{\mathcal{D}}(\mathcal{A}^\#[k_1], \mathcal{A}^\#[k_2]) = \text{Hom}_{\mathcal{D}}(\mathcal{A}^\#, \mathcal{A}^\#[k_2 - k_1]) = 0$$

since $\mathcal{A}^\# \subseteq \mathcal{D}^{\leq 0}$, $\mathcal{A}^\#[k_2 - k_1] \subseteq \mathcal{D}^{\geq 1}$. Now, we get the filtration as follows. Take $\tau^{\leq n} E$, and by (iii) in the first definition, we get a distinguished triangle (now naming the objects with the truncation functors)

$$\tau^{\leq n-1} \tau^{\leq n} E \rightarrow \tau^{\leq n} E \rightarrow \tau^{\geq n} \tau^{\leq n} E \rightarrow (\tau^{\leq n-1} \tau^{\leq n} E)[1]$$

and we show $\tau^{\leq n-1} \tau^{\leq n} E = \tau^{\leq n-1} E$, so if we call the terms E^{n-1}, E^n, A^n respectively, we get our desired triangle with $A^n \in \mathcal{A}^\#[-n]$. The fact that $\tau^{\leq n-1} \tau^{\leq n} E = \tau^{\leq n-1} E$ is essentially the fact that $i^n i^{n-1} = i^n$, i.e., we argue by Yoneda

$$\begin{aligned} \text{Hom}(A, \tau^{\leq n-1} \tau^{\leq n} E) &\cong \text{Hom}(i^{\leq n-1} A, \tau^{\leq n} E) \\ &\cong \text{Hom}(i^{\leq n} i^{\leq n-1} A, E) \\ &\cong \text{Hom}(i^{\leq n} A, E) \\ &\cong \text{Hom}(A, \tau^{\leq n} E). \end{aligned}$$

Now, the bounded assumption will tell us that only finitely many $A_n \neq 0$, and we use these sorts of distinguished with that fact to get the finite filtration.

Now, conversely, let $\mathcal{A}^\#$ be a bounded t-structure as in the second definition, and we show $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ satisfies the first. Essentially by construction, we automatically have (ii). To get (i), we can reformulate our definition of $\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0}$ as

$$\begin{aligned} \mathcal{D}^{\leq 0} &= \langle \mathcal{A}^\#[k_1] \mid k_1 \geq 0 \rangle \\ \mathcal{D}^{\geq 1} &= \langle \mathcal{A}^\#[k_2] \mid k_2 \leq -1 \rangle. \end{aligned}$$

By assumption, we have for $k_1 \geq 0 > -1 \geq k_2$,

$$\text{Hom}(\mathcal{A}^\#[k_1], \mathcal{A}^\#[k_2]),$$

which implies $\mathcal{A}^\#[k_1] \subseteq {}^\perp \mathcal{A}^\#[k_2]$ for all $k_1 \geq 0$, so

$$\mathcal{D}^{\leq 0} = \langle \mathcal{A}^\#[k_1] \mid k_1 \geq 0 \rangle \subseteq {}^\perp \mathcal{A}^\#[k_2],$$

which implies for all $k_2 \leq -1$,

$$\mathcal{A}^\#[k_2] \subseteq (\mathcal{D}^{\leq 0})^\perp,$$

so

$$\mathcal{D}^{\geq 1} = \langle \mathcal{A}^\#[k_2] \mid k_2 \leq -1 \rangle \subseteq (\mathcal{D}^{\leq 0})^\perp,$$

so indeed

$$\text{Hom}(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 1}).$$

To get (iii), take a cohomology filtration, re-indexed so that $A^i \in \mathcal{A}^\#[-i]$ (allowing negative indices and maybe $A_i = 0$),

$$\begin{array}{ccccccc}
0 = E^{-n} & \longrightarrow & E^{-n+1} & \longrightarrow & \cdots & \longrightarrow & E^{-1} & \longrightarrow & E^0 & \longrightarrow & \cdots & \longrightarrow & E^{n-1} & \longrightarrow & E^n = E \\
& \nwarrow & \swarrow & & & & \nwarrow & \swarrow & & & & & \nwarrow & \swarrow & \\
& & A^{-n} & & & & & A^0 & & & & & & A^n &
\end{array}$$

and complete $E^0 \rightarrow E$ to a distinguished triangle

$$E^0 \rightarrow E \rightarrow E/E^0 \rightarrow E^0[1].$$

By construction $E^0 \in \mathcal{D}^{\leq 0}$, and as we have argued before, $E/E^0 \in \mathcal{D}^{\geq 1}$. $\square$

Remark 1.2.9. Let $\mathcal{D}$ be a triangulated category. Given the heart of a bounded t-structure $\mathcal{A}^\sharp$ which induces $(\tau^{\leq 0}, \tau^{\geq 0})$, we had defined for $E \in \mathcal{D}$

$$H_{\mathcal{P}}^i(E) = \tau^{\leq i} \tau^{\geq i} E \in \mathcal{A}^\sharp[-i].$$

We can formulate this differently. If $E \in \mathcal{D}^{\leq n} \cap \mathcal{D}^{\geq -n}$, then it has t-structure cohomology filtration

$$\begin{array}{ccccccc}
0 = E^{-n} & \longrightarrow & E^{-n+1} & \longrightarrow & \cdots & \longrightarrow & E^{-1} & \longrightarrow & E^0 & \longrightarrow & \cdots & \longrightarrow & E^{n-1} & \longrightarrow & E^n = E \\
& \nwarrow & \swarrow & & & & \nwarrow & \swarrow & & & & & \nwarrow & \swarrow & \\
& & A^{-n} & & & & & A^0 & & & & & & A^n &
\end{array}$$

by choosing $E^i = \tau^{\leq i} E$, and then $A^i \in \mathcal{A}^\sharp[-i]$. We exactly then have

$$H_{\mathcal{P}}^i(E) = A^i[i].$$

This will tell us that we can read off the t-structure cohomology in the t-structure cohomology filtration given by a bounded t-structure—all cohomologies are 0, except for those appearing as cones in the filtration.

We then denote this t-structure cohomology by

$$H_{\mathcal{A}^\sharp}^i = H_{\mathcal{P}}^i,$$

as we will later be discussing multiple t-structures at the same time.

Example 1.2.10. standard t-structure on $D(\mathcal{A})$

1.2.3 Hearts are Abelian

Theorem 1.2.11. $\mathcal{A} = \mathcal{D}^{\leq 0} \cap \mathcal{D}^{\geq 0}$ is an abelian category.

Example 1.2.12. To motivate the proof, let's try to use the standard t-structure/truncations to find kernels/cokernels for the standard heart of the derived category.

That is, let $\mathcal{A}$ be an abelian category, $D(\mathcal{A})$ the derived category, and equip $D(\mathcal{A})$ with the standard t-structure $(D(\mathcal{A})^{\leq 0}, D(\mathcal{A})^{\geq 0})$, which has as heart $\mathcal{A}$ (realized as the degree 0 complexes).

Given $f : A \rightarrow B$, we form the cone $C(f)$ as the complex

$$0 \rightarrow A \rightarrow B \rightarrow 0,$$

with A in degree -1 , B in degree 0. Then, we get

$$\begin{aligned}
\tau_{\leq -1} C(f) &= (0 \rightarrow \ker f \rightarrow 0 \rightarrow 0) \\
\tau_{\geq 0} C(f) &= (0 \rightarrow 0 \rightarrow \operatorname{coker} f \rightarrow 0),
\end{aligned}$$

so we see $(\tau_{\leq -1} C(f))[1]$ (shifted to be a degree 0 complex) recovers the kernel, and $\tau_{\geq 0} C(f)$ recovers the cokernel.

We will see that this works for any t-structure.

Proof. This is an additive category, since it is the intersection of two additive subcategories. So, we just have to show that it admits kernels, cokernels, and a morphism decomposition making image canonically isomorphic to coimage.

Let $f : X \rightarrow Y$ be a morphism in $\mathcal{A}$, denote Z its cone, and define

$$K = \tau_{\leq -1} Z \quad C = \tau_{\geq 0} Z.$$

Define k as the composition $\tau_{\leq -1} Z \rightarrow Z \rightarrow X[1]$, and c as the composition $Y \rightarrow Z \rightarrow \tau_{\geq 0} Z$. We claim that $(K[-1], k[-1])$ and (C, c) are the kernel and cokernel of f .

Let's show (C, c) is the cokernel, the kernel is showed similarly. By construction $C \in \mathcal{D}^{\geq 0}$, and from the triangle

$$Y \rightarrow Z \rightarrow X[1] \rightarrow Y[1]$$

where $Y, X[1] \in \mathcal{D}^{\leq 0}$, we need $Z \in \mathcal{D}^{\leq 0}$. □

1.2.4 Tilting at a Torsion Pair

One large source of examples of t-structures is by what is called “tilting at a torsion pair”.

Definition 1.2.13. Let $\mathcal{A}$ be an abelian category. A **torsion pair** $(\mathcal{T}, \mathcal{F})$ is a pair of (strictly) full subcategories of $\mathcal{A}$ satisfying

- (i) $\text{Hom}(\mathcal{T}, \mathcal{F}) = 0$
- (ii) Every object $A \in \mathcal{A}$ fits in a short exact sequence

$$0 \rightarrow T \rightarrow A \rightarrow F \rightarrow 0$$

for some $T \in \mathcal{T}$ and $F \in \mathcal{F}$.

Now, we explain how to tilt at a torsion pair.

Theorem/Definition 1.2.1. Let $\mathcal{D}$ be a triangulated category, and let $\mathcal{A}$ be the heart of a bounded t-structure on $\mathcal{D}$. Let $(\mathcal{T}, \mathcal{F})$ be a torsion pair on $\mathcal{A}$. Then, the following is the heart of a bounded t-structure on $\mathcal{D}$:

$$\mathcal{A}^\sharp = \langle \mathcal{F}[1], \mathcal{T} \rangle = \{X \in \mathcal{D} \mid H_{\mathcal{A}}^0(X) \in \mathcal{T}, H_{\mathcal{A}}^{-1}(X) \in \mathcal{F}, H_{\mathcal{A}}^i(X) = 0 \text{ otherwise}\}.$$

Proof. We give a sketch.

First, $\text{Hom}(\mathcal{A}^\sharp, \mathcal{A}^\sharp[i]) = 0$ for $i < 0$ because

$$\text{Hom}(\mathcal{F}[1], \mathcal{F}[1+i]) = \text{Hom}(\mathcal{F}[1], \mathcal{T}[i]) = \text{Hom}(\mathcal{T}, \mathcal{T}[i]) = \text{Hom}(\mathcal{T}, \mathcal{F}[1+i]) = 0,$$

(where the last equality comes from the definition of a torsion pair) imply

$$\text{Hom}(\langle \mathcal{F}[1], \mathcal{T} \rangle, \langle \mathcal{F}[1], \mathcal{T} \rangle[i]) = 0.$$

Now, we need to produce t-structure cohomology filtrations for $\mathcal{A}^\sharp$. Let $E \in \mathcal{D}$. Our bounded t-structure $\mathcal{A}$ gives us a filtration

$$0 = E^{-n} \longrightarrow E^{-n+1} \rightarrow \dots \rightarrow E^{-1} \longrightarrow E^0 \rightarrow \dots \rightarrow E^{n-1} \longrightarrow E^n = E$$

where $A^i \in \mathcal{A}[-i]$. Then, we “refine” the filtration using the torsion pair as follows. By the torsion pair, we have a short exact sequence

$$0 \rightarrow T^i \rightarrow A^i \rightarrow F^i \rightarrow 0,$$

which will give us a distinguished triangle in $\mathcal{A}$. Thinking of E^i as an extension of A^i by E^{i-1} , we will lift come up with $\tilde{T}^i$ which is an extension of T^i by E^{i-1} , such that we have distinguished triangles.

$$\begin{array}{ccccc}
E^{i-1} & \longrightarrow & \tilde{T}^i & \longrightarrow & E^i \\
& \nwarrow \text{dashed} & \swarrow & \nwarrow \text{dashed} & \swarrow \\
& & T^i & & F^i
\end{array}$$

We can do this by defining

$$\tilde{T}^i = C(T^i[-1] \rightarrow E^{i-1}),$$

and then getting a map $\tilde{T}^i \rightarrow E^i$ by pulling back $E^{i-1} \rightarrow E^i$, after noticing that the composition $T^i[-1] \rightarrow E^{i-1} \rightarrow \tilde{T}^i$ is 0 (and using that Hom is a cohomological functor). Then, the cone over $E^{i-1} \rightarrow \tilde{T}^i$ is T^i by definition, and the cone over $\tilde{T}^i \rightarrow E^i$ is F^i by the octahedral axiom (i.e., analyze the composition $E^{i-1} \rightarrow \tilde{T}^i \rightarrow E^i$).

The point now is to filter E by the $\tilde{T}^i$ instead of the E^i .

Putting two of these diagrams next to each other and deleting the left/right ends, we get the following.

$$\begin{array}{ccccc}
\tilde{T}^{i-1} & \longrightarrow & E^{i-1} & \longrightarrow & \tilde{T}^i \\
& \nwarrow \text{dashed} & \swarrow & \nwarrow \text{dashed} & \swarrow \\
& & F^{i-1} & & T^i
\end{array}$$

We now need to compute the cone over $\tilde{T}^{i-1} \rightarrow \tilde{T}^i$, which we denote by $\tilde{A}^i$. Applying the octahedral axiom, we get a distinguished triangle

$$F^{i-1} \rightarrow \tilde{A}^i \rightarrow T^i \rightarrow F^{i-1}[1].$$

Taking $\mathcal{A}$ cohomology, and recognizing that

$$H_{\mathcal{A}}^j(F^{i-1}) = \begin{cases} F^{i-1}[i-1] & j = i-1 \\ 0 & \text{otherwise} \end{cases}, \quad H_{\mathcal{A}}^j(T^i) = \begin{cases} T^i[i] & j = i \\ 0 & \text{otherwise} \end{cases}$$

we then get exact sequences

$$0 \rightarrow F^{i-1}[i-1] \rightarrow H_{\mathcal{A}}^{i-1}(\tilde{A}^i) \rightarrow 0$$

and

$$0 \rightarrow H_{\mathcal{A}}^i(\tilde{A}^i) \rightarrow T^i[i] \rightarrow 0,$$

so in other words

$$\begin{aligned}
H_{\mathcal{A}}^{-1}(\tilde{A}^i[i]) &= F^{i-1}[i-1] \in \mathcal{F} \\
H_{\mathcal{A}}^0(\tilde{A}^i[i]) &= T^i[i] \in \mathcal{T},
\end{aligned}$$

so indeed

$$\tilde{A}^i \in \mathcal{A}^{\sharp}[-i],$$

so we get an $\mathcal{A}^{\sharp}$ t-structure cohomology filtration as desired. $\square$

Tilting provides tons of examples of t-structures—starting with the standard t-structure $\mathcal{A} \subseteq D^b(\mathcal{A})$, we can tilt over and over again to find more and more t-structures. One question we can ask is whether two given t-structures are tilts of each other. We can use the following proposition to help us answer this questions.

Proposition 1.2.14. Let $\mathcal{A}^{\sharp}, \mathcal{A}$ be hearts of bounded t-structures on a triangulated category $\mathcal{D}$, and assume

$$\mathcal{A}^{\sharp} \subseteq \langle \mathcal{A}[1], \mathcal{A} \rangle.$$

Then, $\mathcal{A}^{\sharp}$ is a tilt of $\mathcal{A}$ at the torsion pair

$$\mathcal{T} = \mathcal{A} \cap \mathcal{A}^{\sharp}, \quad \mathcal{F} = \mathcal{A} \cap \mathcal{A}^{\sharp}[-1].$$

Proof. It's clear that

$$\mathrm{Hom}(\mathcal{T}, \mathcal{F}) = 0,$$

since $\mathcal{T} \subseteq \mathcal{A}^\#$ and $\mathcal{F} \subseteq \mathcal{A}^\#[-1]$.

Now, we need to show that every object $A \in \mathcal{A}$ is an extension of $\mathcal{F}$ by $\mathcal{T}$. To do this, we study the $\mathcal{A}^\#$ cohomology filtration of A , which will be something like

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & A^{-2} & \longrightarrow & A^{-1} & \longrightarrow & A^0 & \longrightarrow & A^1 & \longrightarrow & A^2 & \longrightarrow & \cdots \\ & & & \swarrow & \searrow & & \swarrow & \searrow & \swarrow & \searrow & & & \\ & & & & H^{-1} & & & & H^0 & & & & \\ & & & \swarrow & \searrow & & \swarrow & \searrow & \swarrow & \searrow & & & \\ & & & & & & H^1 & & & & & & \\ & & & & & & & & H^2 & & & & \end{array}$$

and we want to argue that only H^0, H^1 are nonzero, so that we get a distinguished triangle

$$H^0 \rightarrow A \rightarrow H^1 \rightarrow H^0[1],$$

FINISH THIS!!

□

1.2.5 Examples in Algebraic Geometry

Example 1.2.15. Let (X, H) be a polarized surface. We will define a tilt of $D^b(X)$ which will be used later to define a stability condition.

Let $\mu \in (-\infty, \infty]$. Then, define

$$\mathcal{T}_\mu = \{\text{torsion sheaves}\} \cup \{\text{sheaves with } \mu^- > \mu\}$$